JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"
Session: 2023-27

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

B. Tech.

In

Computer Science and Engineering

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3rd Semester Computer Science & Engineering

Sr. No.	Category of Subject	Course Code	Course Name	Teaching Scheme			Evaluation Scheme				
				L	T/A	P	CA	MSE	ESE	Total	Credit
1	PCC	CS3T001	Computer Networks	3	0	0	20	20	60	100	3
2	PCC	CS3T002	Operating Systems	3	0	0	20	20	60	100	3
3	PCC	CS3T003	Computer Architecture & Organization	3	0	0	20	20	60	100	3
4	MDM	CS3M001	Numerical Technique & Discrete Mathematics	2	0	0	20	20	60	100	2
5	OEC	CS3O001	Open Elective -I	3	0	0	20	20	60	100	3
6	EEMC	CS3H001	Entrepreneurship Development	2	0	0	20	20	60	100	2
7	VEC	CS3V001	UHV-II	2	0	0	20	20	60	100	2
8	PCC	CS3L005	Object Techniques & UML Lab	0	0	2	60	-	40	100	1
9	PCC	CS3L006	Python Programming Lab	0	0	2	60	-	40	100	1
10	PCC	CS3L007	Web Technology Lab	0	0	2	60	-	40	100	1
11	CEP/FP	CS3F001	Community Engagement Project	0	0	4	60	-	40	100	2
12											
				18	**0**	**10**	**380**	**140**	**580**	**1100**	**23**

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3T001**	**Computer Network**	3	0	0	3

Prerequisites for the course	
1.	Basics of Computer Devices & Connecting Devices.
2.	Fundamentals of Protocols, Data & Signals.

Prior Reading Material/useful links	
1.	https://onlinecourses.nptel.ac.in/noc23_cs35/preview
2.	https://www.capectc.org/o/ctc/page/computer-networking-security

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Student will be able to describe the basics of network and its hardware components
2	CO2	Students will be able to explain the different network models.
3	CO3	Students will be able to interpret the various functions and protocols of network models.
4	CO4	Students will be able to assess different transmission media with its connectors.
5	CO5	Students will be able to summarize the concepts of network security and privacy.

Syllabus:

Course Contents		Hours
Unit I	**Introduction to Data Communication and Computer Networks:** Definition, Characteristics, Components, Data & Signal, Types of Data & signals, Data Representation, Need of Computer Networks, Advantages and Disadvantages, Goals and Application of Computer Network, Network Hardware Components, Physical Topology, Types of Network, PAN, LAN, MAN, WAN.	08
Unit II	**Physical Layer:**, Transmission Mode, Transmission Impairment, Data Rate Limits, Nyquist Bit Rate, Transmission Media, Switching Techniques. **Data Link Layer:** Framing Methods, Error Detection and Correction Methods, MAC layer multiple access protocols	07

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	(CSMA,CSMA/CD,CSMA/CA), Channelization(FDMA, TDMA, CDMA), Introduction to Virtual LAN. IP Address.	
Unit III	**Network Layer:** Design Issues, Store and Forward Packet, Switching Connection Less and Connection Oriented Services, Routing Algorithms, Shortest Path, Flooding, Distance Vector Routing, Hierarchical Routing, Congestion Control Algorithms. OSI Reference Model & Architecture, TCP/IP Reference Model.	08
Unit IV	**Transport Layer:** Elements of Transport Protocols, Connection Establishment and Termination, TCP Header Format, UDP Header Format, Sliding Window, Timers, Retransmission. **Session and Presentation Layer:** Session Layer Design Issues, Responsibilities of Presentation Layer.	07
Unit V	**Application Layer and Network Security:** Responsibilities of Application Layer, Application Layer Services (DNS, E-mail, MIME, SMTP, FTP, TFTP),Architecture of WWW and HTTP, Introduction to Cryptography, Security Services, Introduction to Symmetric and Asymmetric Key Cryptography, Digital Signature.	06

Text Books	
1.	Data Communications and Networking - Behrouz A. Forouzan, Fifth Edition TMH, 2013.
2.	Computer Networks - Andrew S Tanenbaum, 4th Edition, Pearson Education.
3.	Kurose and Ross, "Computer Networking - A top-down approach", Seventh Edition, Pearson, 2017.
4.	Peterson and Davie, "Computer Networks, A Systems Approach", 5th ed., Elsevier,

Reference Books	
1.	An Engineering Approach to Computer Networks - S. Keshav, 2nd Edition, Pearson Education.
2.	Understanding communications and Networks, 3rd Edition, W. A. Shay, Cengage Learning.

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc21_cs18/preview
2.	https://onlinecourses.nptel.ac.in/noc19_cs75/preview

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Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3T002**	**Operating System**	3	0	0	**3**

Prerequisites for the course	
1.	Basics Knowledge of Operating System.
2.	Fundamental services provided by and the design of an operating system.
3.	Introduction of different approaches to memory management

Prior Reading Material/useful links	
1.	https://www.tutorialspoint.com/operating_system/os_overview.htm
2.	https://www.tutorialspoint.com/computer_fundamentals/computer_operating_system.htm
3.	https://www.mygreatlearning.com/blog/what-is-operating-system/

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Understand the basics of how operating systems work.
2	CO2	Students will be able to Explain how processes and CPU scheduling function in an operating system.
3	CO3	Students will be able to Solve common process synchronization problems.
4	CO4	Students will be able to Describe memory management concepts, including virtual memory.
5	CO5	Students will be able to Comprehend disk management and the role of file systems in an operating system.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Concept of Operating System, User View, System View, Computer System Organization, Bootstrap Program, Storage Structure, Types of Operating Systems. **Operating System Structure:** Monolithic, Layered, Micro Kernel, Operating System Services, User and Operating System Interfaces:	7

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	Command Interpreters, Graphical User Interface. System Calls, Types of System Call, System Programs.	
Unit II	**Process Concept:** Process Definition, Process in Memory, Process State, Process Control Block(PCB), Operation on Process, Context Switching. **Threads:** Definition, Benefits of Threads, Types of Threads, Different State of Thread. **Process Scheduling:** Scheduling Objective, CPU – I/O Burst Cycle, CPU Scheduler: Types of Scheduler, Scheduling Criteria. **Scheduling Algorithms:** Pre-Emptive and Non Pre-emptive, FCFS, SRTF, Priority, RR.	8
Unit III	**Synchronization:** Critical Section Problem, Race Condition, Peterson Solution, Semaphores. Classic Problem IPC problem: Producer Consumer Problem, Reader Writer Problem. The Dinning-Philosophers Problem. **Deadlocks:** System Model, Deadlock Characterization, Methods of Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance: Banker's algorithm, Deadlock Detection and Recovery	7
Unit IV	**Memory Management:** Basic Concepts, Logical and Physical Address Mapping, Swapping **Memory Allocation**: Contiguous Memory Allocation, Fixed Partition and Variable Partition, **Fragmentation:** Internal and External Fragmentation **Paging:** Basic method , Paging Model for Logical and Physical Memory , Paging Hardware with TLB, Advantage and Disadvantage of Paging.	7
Unit V	**Virtual Memory Management:** Basic of Virtual Memory, Demand Paging, Page Replacement Algorithm: FIFO, LRU, Optimal **Disk Management:** Disk Structure, Disk Scheduling– FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK **File System:** File concepts, File Attributes, File Operations, File Types. File Access Method : Sequential Access ,Direct Access	7

Text Books	
1.	Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, Wiley Publication, 8th Edition, 2008.
2.	Andrew S. Tanenbaum, Modern Operating System, PHI Publication, 4thEdition, 2015
3.	Richard Stevens, Stephen Rago, Advanced Programming in the UNIX Environment, Pearson Education,
4.	D. M. Dhamdhere, Systems Programming and Operating Systems, McGraw-Hill, 2ndEdition, 1996.

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Reference Books	
1.	Garry Nutt, Operating Systems Concepts, Pearson Publication, 3rd Edition, 2003.
2.	Harvey M. Deitel, An Introduction to Operating Systems, Addison-Wesley Publication, 2nd Edition, 1990.
3.	Thomas W. Doeppner, Operating System in Depth: Design and Programming, Wiley
4.	M. J. Bach. Design of the Unix Operating System, Prentice Hall of India, 1986.Publication, 2011.

Useful Links	
1.	https://www.google.co.in/books/edition/Introduction_to_Operating_System_Design/KisFwig6YnAC?hl=en&gbpv=1&dq=Operating+System&printsec=frontcover
2.	https://www.google.co.in/books/edition/Operating_System_Concepts/hIdQAAAAMAAJ?hl=en&gbpv=1&bsq=Operating+System&dq=Operating+System&printsec=frontcover
3.	https://www.google.co.in/books/edition/Operating_System/Ad9qKEaVzQIC?hl=en&gbpv=1&dq=Operating+System&printsec=frontcover
4.	https://www.google.co.in/books/edition/Operating_Systems/gSy8qdmf3qEC?hl=en&gbpv=1&dq=Operating+System&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3T003**	**Computer Architecture & Organization**	3	0	0	**3**

Prerequisites for the course	
1.	Basics of Digital Electronics and Microprocessor.
2.	Fundamentals of Number Systems.

Prior Reading Material/useful links	
1.	https://www.youtube.com/watch?v=Ol8D69VKX2k
2.	https://www.youtube.com/watch?v=4TzMyXmzL8M
3.	https://www.youtube.com/watch?v=leWKvuZVUE8

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Summarize the organization and operation of digital computers.
2	CO2	Students will be able to Study and apply knowledge of processor instruction sets and its execution.
3	CO3	Students will be able to Demonstrate computer arithmetic operations on integer and floating-point numbers..
4	CO4	Students will be able to Describe the organization of memory system.
5	CO5	Students will be able to Explain concepts of I/O organization and pipelining of a processor.

Syllabus:

Course Contents		Hours
Unit I	**Basic Structure of Computer:** Functional Units, Architecture of a small accumulator based CPU, A typical CPU with general register organization, Instruction execution cycle, Addressing modes, Instruction Format. **Processing Unit:** Execution of a complete instruction, Sequencing of control Signals, types of Buses, Single, Two, multiple bus structure.	08

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Unit II	**Computer Arithmetic:** Binary Addition, Addition and subtraction, Multiplication of unsigned binary integers, Booth's algorithm for Two's complement multiplication unsigned, Unsigned binary division, IEEE Floating-Point representation, Floating Point arithmetic.	07
Unit III	**Control Unit:** Control Unit operation: Introduction, Micro-operations, Control of the Processor, Hardwired implementation. **Micro Programmed Control:** Microinstruction formats, Micro programmed control unit, Functioning of micro programmed control unit, Microinstruction sequencing techniques.	07
Unit IV	**The Memory System:** Internal organization of memory chip, Static memories, Dynamic RAMs, Read-Only Memories, Memory interleaving, Cache Memory, Mapping techniques, Virtual memory, Memory Management requirements, I/O modules, Programmed I/O, Interrupt Driven I/O, DMA.	07
Unit V	**Pipelining and parallel Processing:** Pipelining: Introduction, Pipeline organization, Pipelining issues, Memory delays, Branch delays, Parallel Processing, Types of parallel processor systems, Vector processing **Processors:** RISC & CISC Processors, Pentium processor, superscalar processor.	07

Text Books	
1.	V. Carl Hamacher& S. Zaky: Computer Organization, Fourth Edition, McGraw-Hill (ISE).
2.	Computer Organization and Design David A.Patterson & John L. Hennessy Morg, Fourth Edition, McGraw-Hill (ISE).

Reference Books	
1.	Stallings. W: Computer Organization &Arcitecture, Fifth Edition, Pearson Education.
2.	Tananbaum A. S: Structured Computer Organization, Fifth Edition, Pearson Education

Useful Links	
1.	https://onlinecourses.nptel.ac.in/noc22_cs88/preview
2.	https://nptel.ac.in/courses/106106166

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Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3M001**	**Numerical Technique & Discrete Mathematics**	2	0	0	**2**

Prerequisites for the course	
1.	Basics concept of Numerical method tool to solve the problems in real life
2.	understand the application of Mathematics in engineering and in real life.

Prior Reading Material/useful links	
1.	https://www.geeksforgeeks.org/matrices/
2.	https://people.umass.edu/~prokofev/P605/complex_functions.pdf

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able Describe the concept of Numerical method, Bisection method, Newton-Raphson method and their convergences, Gauss elimination method, Crout's method.
2	CO2	Students will be able to Illustrate the concept of Mathematical Expectation, Variance and standard deviation, Moments and moment generating function, Characheteristic function, Conditional Expectation, Variance and moments.
3	CO3	Students will be able to Apply the concept of Joint Probability and Random Variable, Probability Function of CRV, Joint Probability Function of DRV,Join Distribution Function of DRV.
4	CO4	Students will be able to Evaluate the problem base on the concept of Propositions and Logical Operation, Quantifiers, Fourier transforms, complex variables, Numerical Linear Algebra.
5	CO5	Students will be able to Create the new concept by using the theory of Laplace Transform, Inverse Laplace Transform, Fourier transform, complex variables, Numerical Linear Algebra.
6	CO6	Students will be able to Create the new concept by using the theory of Group Theory, Group (only definition and examples) Subgroups and Homomorphism Cosets and Lagrange's theorem, Normal subgroups.

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Syllabus:

Course Contents		Hours
Unit I	**Linear Algebra:** Linear dependence and independence of vectors, consistency of homogenous and non-homogenous equations, Characteristics equation, Eigen values and Eigen vectors, Sylvester's theorem [without proof],Quadratic form expressed in matrices, linear transformation of quadratic form (Canonical form), differentiation and integration of matrices.	04
Unit II	**Numerical Methods:** Error analysis, Solutions of algebraic and transcendental equations. Bisection method, Newton-Raphson method and their convergences. Solution of system of linear equations by Gauss Seidel method, Gauss elimination method, Crout's method.	05
Unit III	**Mathematical Expectation:** Mathematical Expectation, Variance and standard deviation, Moments and moment generating function, Characteristic function, Expectation and Variance for joint distribution and covariance, Conditional Expectation, Variance and moments.	05
Unit IV	**Joint Probability and Random Variable:** Joint Distribution of Discreet Random Variable: Joint Probability Function of DRV, Join Distribution Function of DRV ,Marginal Probability Function of DRV, Joint Distribution of Continuous Random Variable: Joint Probability Function of CRV, Join Distribution Function of CRV ,Marginal Probability Function of CRV.	04
Unit V	**Group Theory:** Binary Operations , Properties , Semigroups ,Monoids, subsemigroup, Submonoid, Isomorphism and Homomorphism, Group (only definition and examples) Subgroups and Homomorphism Cosets and Lagrange's theorem, Normal subgroups.	06

Text Books	
1.	Higher Engineering Mathematics by B. S. Grewal, Khanna Publishers, New Delhi.
2.	Advanced Engineering Mathematics by Erwin Kreyszig, John Wiley & Sons, New York.
3.	A Course in Engineering Mathematics (Vol III) by Dr. B. B. Singh, Synergy Knowledgeware, Mumbai.
4.	A Text Book of Applied Mathematics (Vol I & II) by P. N. Wartikar and J. N. Wartikar, Pune Vidyarthi Griha Prakashan, Pune.

Reference Books	
1.	Higher Engineering Mathematics by B. V. Ramana, Tata McGraw-Hill Publications, NewDelhi.
2.	A Text Book of Engineering Mathematics by Peter O' Neil, Thomson Asia Pte Ltd.,Singapore.
3.	Advanced Engineering Mathematics by C. R. Wylie & L. C. Barrett, Tata Mcgraw-Hill Publishing Company Ltd., New Delhi.

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Useful Links	
1.	https://byjus.com/jee/matrices/
2.	https://betterexplained.com/articles/an-interactive-guide-to-the-fourier-transform/

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Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3O001**	**Open Elective-I (Cyber Law & Cyber Crime)**	3	0	0	**3**

Prerequisites for the course	
1.	Introduction to cryptography.
2.	Some basic of cyber security.

Prior Reading Material/useful links	
1.	https://www.stu.edu/law/academics/cybersecurity-law/learningoutcomes/
2.	https://www.careers360.com/courses/cyber-law-course

Course Outcomes:

Sr. No	Course Outcome Number	CO statement
1	CO1	Students will be able to Understand the different types of cyber security threats posed to computer systems and networks
2	CO2	Students will be able to Identify national security implications from threats in the cyber domain.
3	CO3	Students will be able to Apply the legal authorities and obligations of government and the private sector to protect computer systems and networks
4	CO4	Students will be able to analyse the national security policy decisions, directives, and actions for developing and implementing cyber security policy in relation to federal laws, executive orders, regulations, and on-going cases.
5	CO5	Students will be able to evaluate and communicate the human role in security systems with an emphasis on ethics.

Syllabus:

Course Contents		Hours
Unit I	**Introduction:** Understanding Cyber Crimes and Cyber Offences, Crime in context of Internet, Types of Crime in Internet, **Crimes targeting Computers:** Definition of Cyber Crime & Computer related Crime. History, Development and Reasons for Growth of Cyber Crimes.	08

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Unit II	**Role in Cyber & Cyber Crimes:** Social Media and its Role in Cyber World, Fake News, Defamation, Online Advertising, Prevention of Cyber Crimes & Frauds, Critical analysis & loop holes of The IT Act, 2000 in terms of cyber-crimes, **Cyber Crimes:** Freedom of speech in cyber space & human right issues, International position on Free Speech in Internet.	07
Unit III	**Indian Penal Law and Cyber Crimes:** Fraud, Hacking, Mischief, Trespass, Defamation, Stalking, Spam, **Issues of Internet Governance:** Freedom of Expression in Internet, Issues of Censorship, Hate Speech, Sedition, Libel, Subversion, Privacy.	07
Unit IV	**Cyber Law:** International Perspective, EDI: Concept and legal Issues, UNCITRAL Model Law, Electronic Signature Laws of Major Countries Cryptography Laws, Cyber Laws of Major Countries, EU Convention on Cyber Crime	07
Unit V	**International Organizations and Their Roles:** ICANN, URDP, WTO and TRIPS, Interpol & Europol, Impact of Cyber warfare on Privacy Identity, Net Neutrality and EU Electronic communication Regulatory framework, WCAG, Social Networking sites Vis – a – Vis Human Right.	07

Text Books	
1.	Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd
2.	Information Warfare and Security by Dorothy F. Denning, Addison Wesley.
3.	Security in the Digital Age: Social Media Security Threats and Vulnerabilities by Henry A. Oliver, Create Space Independent Publishing Platform

Reference Books	
1.	Prashant Mali: Cyber Law & Cyber Crimes Sumplified, Cyber Infonedia Publishe
2.	Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., Latest Edition

Useful Links	
1.	https://lawbhoomi.com/cyber-law-notes/
2.	https://www.upcounsel.com/cyber-law

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Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3H001**	**Entrepreneurship Development**	2	0	0	**2**

Prerequisites for the course	
1.	Basics of Universal Human Value and entrepreneurial behaviour.
2.	Knowledge of the social, ethical and culture and Consumer Affairs

Prior Reading Material/useful links	
1	https://www.google.co.in/books/edition/Entrepreneurship_and_Economic_Developmen/deeFDAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover
2	https://www.google.co.in/books/edition/Innovation_and_Entrepreneurship/OiuDBAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Identify and validate of ideas.
2	CO2	Students will be able to Remember Patent registration of Innovation.
3	CO3	Students will be able to Understand roles and responsibilities of Entrepreneurship.

Syllabus:

Course Contents		Hours
Unit I	**Innovation:** Concept of creativity, innovation, invention, discovery. Methods for development of creativity, convergent & divergent thinking etc. Introduction to Intellectual Property Rights (IPR), Patent and laws related to patents.	07
Unit II	**Entrepreneurship**: Theories of Entrepreneurship: Concept of entrepreneurship, its relations in economic developments, Eventuation of concept of entrepreneur, characteristics of an Entrepreneur, Types of entrepreneurs, Qualities of entrepreneur, Factors affecting growth of entrepreneurship.	07
Unit III	**Role of Entrepreneurial Bodies:** Theory of achievement, motivation, Medelland's. experiment, Women entrepreneurship, Role of SSI, it's advantages & limitations, policies governing small scale industries,	07

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	Procedure to set up small scale industrial unit, Advantages and limitations of SSI.	
Unit IV	**Role of Entrepreneurial Support:** Factors governing project selection, Market survey, Preparation of project report. Financial, technical & market analysis of project. Entrepreneurial support systems, Role of consultancy organization like, District Industrial Centre, State Industrial Development Corporation, Financial institution, Latest SSI schemes of DIC (to be confirmed from DIC from time to time.	07

Text Books	
1.	Entrepreneurship Development, S. S. Khanka, S. Chand Publishers
2.	Creativity Innovation & Entrepreneurship, Zechariah James Blanchard, Needle Rat Business, Publishers.

Reference Books	
1.	Khanna, S. S., Entrepreneurial Development, S. Chand, New Delhi.
2.	Hisrich D. Robert, Michael P. Peters, Dean A. Sheperd, Entrepreneurship, McGraw-Hill,6 ed.

Useful Links	
1	https://www.google.co.in/books/edition/Creativity_and_Innovation_in_Entrepreneu/f1c7EAAAQBAJ?hl=en&gbpv=1&dq=Innovation+and+Entrepreneurship+Development&printsec=frontcover
2	https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=&cad=rja&uact=8&ved=2ahUKEwj14Jaw9L__AhWgU2wGHaLcCNUQFnoECCYQAQ&url=https%3A%2F%2Fwww.dpgpolytechnic.com%2Fdownloads%2Ffiles%2Fn5c419bcb66c09.pdf&usg=AOvVaw3cbsLd-So8C6Pgj0LW3S35

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3V001**	**Universal Human Value-II**	2	0	0	**2**

Prerequisites for the course	
1.	Basics of Communication Skill.
2.	Fundamental concept of Organization Behaviour

Prior Reading Material/useful links	
1.	https://vvce.ac.in/wp-content/uploads/2021/04/Realising-Aspirations-of-NEP2020-UHV.pdf
2.	https://www.geeksforgeeks.org/human-values-concept-and-importance/
3.	https://www.researchgate.net/publication/320613836_The_importance_and_need_of_universal_human_values_in_engineering_students_life

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to become more aware of their surroundings, society, social problems and their sustainable solutions, while keeping human relationships and human nature in mind.
2	CO2	Students would have better critical ability.
3	CO3	Students would also become sensitive to their commitment towards what they believe in (humane values. Humane relationships and humane society).
4	CO4	Students would be able to apply what they have learnt to their own self in different day-to-day settings in real life, at least a beginning would be made in this direction.

Syllabus:

Course Contents		Hours
Unit I	**Course Introduction** - Need, Basic Guidelines, Content and Process for Value Education, Purpose and motivation for the course, recapitulation from Universal Human Values-I . Self-Exploration–what is it? - Its content and process; 'Natural Acceptance' and Experiential Validation- as the process for self-exploration. Continuous Happiness and Prosperity- A look at basic Human Aspirations . Right understanding, Relationship and Physical Facility- the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly- A critical	7

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	appraisal of the current scenario. Method to fulfil the above human aspirations: understanding and living in harmony at various levels. Include practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as arbitrariness in choice based on liking-disliking.	
Unit II	**Understanding Harmony in the Human Being** - Harmony in Myself! Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - happiness and physical facility. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the characteristics and activities of 'I' and harmony in 'I'. Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensureSanyam and Health. Include practice sessions to discuss the role others have played in making material goods available to me. Identifying from one's own life. Differentiate between prosperity and accumulation. Discuss program for ensuring health vs dealing with disease	7
Unit III	**Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship:**Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship. Understanding the meaning of Trust; Difference between intention and competence. Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals. Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family. Include practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Discuss with scenarios. Elicit examples from students' lives	7
Unit IV	**Understanding Harmony in the Nature and Existence - Whole existence as Coexistence** Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Understanding Existence as Co-existence of mutually interacting units in all-pervasive space.Holistic perception of harmony at all levels of existence. Include practice sessions to discuss human being as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.	7

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Unit V	**Implications of the above Holistic Understanding of Harmony on Professional Ethics:** Natural acceptance of human values. Definitiveness of Ethical Human Conduct. Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people- friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order: a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers b. At the level of society: as mutually enriching institutions and organizations.	7

Text Books	
1.	Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010
2.	JeevanVidya: EkParichaya, ANagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
3.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.

Reference Books	
1.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
2.	Rediscovering India - by Dharampal
3.	Economy of Permanence - J C Kumarappa
4.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi

Useful Links	
1.	https://www.google.co.in/books/edition/Human_Values/XhuMFyzdMT8C?hl=en&gbpv=1&dq=Universal+Human+Values&printsec=frontcover
2.	https://www.google.co.in/books/edition/Human_Values_and_Professional_Ethics/CSDzDwAAQBAJ?hl=en&gbpv=1&dq=Universal+Human+Values&printsec=frontcover

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3L005**	**Object Techniques & UML Lab**	0	0	2	**1**

Prerequisites for the course	
1.	Basics knowledge of Object Oriented Programming.
2.	Fundamental of UML Models, Tools.

Prior Reading Material/useful links	
1.	https://www.cet.edu.in/noticefiles/285_OOPS%20lecture%20notes%20Complete.pdf
2.	https://www.uml-lab.com/
3.	https://www.clear.rice.edu/comp310/Eclipse/UMLLab/

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Programming Basics of OOPs and UML Concepts.
2	CO2	Students will be able to Implement C++ program by using structure, functions and Draw Class & Activity Diagram.
3	CO3	Students will be able to Develop Socket Programing and draw Deployment Model.
4	CO4	Students will be able to Use object-oriented approaches to software problems in C++.
5	CO5	Students will be able develop applications for a range of problems using object-oriented programming techniques and UML Models.

List of Practical's

Course Contents		Hours
0	Introduction of Object Oriented Programming & UML Lab	2
1	WAP to Store Information of Students using Structure also Draw Class Diagram of Students with attributes.	2
2	WAP for ATM machine in C++ using Functions and Draw Activity Diagram for it.	2
3	Draw Use Case Diagram of Library Scenario also Write A C++ Program To Read and Write The Library Data	2
4	Implement client server architecture in C++ and Draw Deployment diagram for Client server system	2

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5	WAP for Timing constraints in real-time system also draw Sequence diagram on it.	2
6	WAP Login and Register System using File handling in C++ and Draw Collaboration diagram for it.	2
7	WAP for User-defined Function in C++ and make a component diagram for my web service	2
8	Implementing a Finite State Machine in C++ and Draw State Transition Diagram.	2
9	Draw Diagram for the relation between a Component & Class.	2
10	Draw Deployment diagram for Physical Connection System.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Object Oriented Programming with C++ by E. Balagurusamy, McGraw-Hill Education (India)
2.	C++ and Object Oriented Programming – Jana, PHI Learning.
3.	The Elements of UML(TM) 2.0 Style by Scott W. Ambler Cambridge University Press, 09-May-2005 - Computers - 188
4.	Fundamentals of Object-Oriented Design in Uml by Meilir Page-Jones, Larry L. Constantin

Reference Books	
1.	Mastering C++ - Venugopal, McGraw-Hill Education (India)
2.	Object Oriented Programming with C++ - Rajiv Sahay, Oxford
3.	Using UML Activities to model Business Processes, A Handbook for Practitioners Ed Walters.

Useful Links	
1.	https://biet.ac.in/pdfs/C++%20LAB%20MANUAL.pdf
2.	http://www.uop.edu.pk/ocontents/good%20startter.pdf
3.	https://www.studocu.com/in/document/adithya-institute-of-technology/computer-science/r19-uml-lab-manual/37364626
4.	https://www.scribd.com/doc/67143168/UML-Lab-Manual

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3L006**	**Python Programming (Lab)**	0	0	2	**1**

Prerequisites for the course	
1.	Basics Knowledge of Computer programming.
2.	Introduction to Computer programming Lab.

Prior Reading Material/useful links	
1.	https://www.rgmcet.edu.in/assets/img/departments/CSE/materials/R19/2-1/Python%20Lab.pdf
2.	https://mrcet.com/pdf/Lab%20Manuals/CSE/(R18A0588)%20Python%20Programming%20%20Lab%20Manual.pdf
3.	https://python-iitk.vlabs.ac.in/

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Write, test, and debug simple Python programs
2	CO2	Students will be able to Implement Python programs with conditionals and loops.
3	CO3	Students will be able to Develop Python programs step-wise by defining functions and calling them
4	CO4	Students will be able to Use Python lists, tuples, dictionaries for representing compound data.
5	CO5	Students will be able to Read and write data from/to files in Python

List of Practical's

Course Contents		Hours
0	Introduction of Python Programming Lab	2
1	WAP to implement stack using arrays.	2
2	WAP to evaluate a given postfix expression using stacks.	2
3	WAP to convert a given infix expression to postfix form using stacks.	2
4	WAP to implement circular queue using arrays.	2
5	WAP to implement double ended queue (de queue) using arrays.	2

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6	WAP to implement a stack using two queues such that the push operation runs in constant time and the pop operation runs in linear time.	2
7	WAP to implement a stack using two queues such that the push operation runs in linear time and the pop operation runs in constant time.	2
8	Write a program to implement a queue using two stacks such that the enqueue operation runs in constant time and dequeue operation runs in linear time.	2
9	Write programs to implement the following data structures: a. Single linked list b. Double linked list	2
10	Implement the following sorting algorithms: a. Insertion sort b. Merge sort c. Quick sort d. Heap sort	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Python for the Lab By Aquiles Carattino · 2020 Lulu.com
2.	Explorations in Computing An Introduction to Computer Science By John S. Conery · 2011 Taylor & Francis
3.	Introduction to Computer Science Using Python A Computational Problem-Solving Focus By Charles Dierbach · 2013 Wiley

Reference Books	
1.	The Absolute Beginner's Guide to Python Programming A Step-by-Step Guide with Examples and Lab Exercises By Kevin Wilson · 2022Apress
2.	Absolute Beginner's Python Programming Full Color Guide with Lab Exercises The Illustrated Guide to Learning Computer Programming By Kevin Wilson · 2022 Elluminet Press

Useful Links	
1.	https://www.google.co.in/books/edition/Python_for_the_Life_Sciences/y5CyDwAAQBAJ?hl=en&gbpv=1&dq=Python+Programming+(Lab)&printsec=frontcover
2.	https://www.mrecacademics.com/DepartmentStudyMaterials/20201223-python%20programming%20lab%20manual.pdf
3.	https://cdlsiet.ac.in/wp-content/uploads/2022/03/PYTHON-Lab-Manual.pdf
4.	https://www.jnit.org/wp-content/uploads/2020/04/Python-Lab-Manual-converted.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3L007**	**Web Technology Lab**	0	0	2	**1**

Prerequisites for the course	
1.	Basics of Internet Programming like coding, designing, Testing.
2.	Fundamental Knowledge of Computing and Programming.

Prior Reading Material/useful links	
1.	https://www.bharathuniv.ac.in/downloads/csc/BCS7L2Web%20Technology%20Lab.pdf
2.	https://gnindia.dronacharya.info/CSE/Downloads/Labmanuals/Web-Tech-Lab Manual.pdf

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will be able to Design web pages using HTML, DHTML and Cascading Styles sheets.
2	CO2	Students will be able to access and Validate form data using JavaScript and PHP.
3	CO3	Students will be able to connect to database using JSP and JDBC and perform various operations.
4	CO4	Students will be able to connect to MySQL using PHP and perform various operations
5	CO5	Students will be able to Build and consume web services.

List of Practical's

Course Contents		Hours
0	Introduction of Web Technology Lab	
1	Design the following static web pages required for an online book store web site. i. Home Page ii. Login Page iii. Catalogue Page	2
2	Design the following static web pages required for an online book store web site. i. Registration Page	2

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	ii. Cart Page	
3	Design a web page using CSS which includes the following: i.Use different font and text styles ii. Set a background image for both the page and single element on the page. iii. Define styles for links iv. Working with layers v. Adding a Customized cursor	2
4	i. Write a JavaScript to validate the fields of the login page. ii. Write a JavaScript to validate the fields of the Registration page	2
5	Write an XML file which will display the Book information which includes the following: Title of the book, Author Name, ISBN number, Publisher name, Edition and Price. Validate the above document using DTD and XML Schema	2
6	i. Write a PHP program to validate the fields of the login page. ii. Write a PHP program to validate the fields of the Registration page	2
7	Write a JSP to connect to the database and extract data from the tables and display them to the user.	2
8	Design a JSP to insert the details of the users who register through the registration page and store the details in to the database.	2
9	Write a PHP program to connect to MySQL database which retrieves the data from the tables and display them to the user.	2
10	Write a PHP program to insert the details entered by the user in the Registration form into MySQL database.	2

Note: 2 Practical based on Content Beyond Syllabus.

Text Books	
1.	Introduction To Web Technology (Paperback) by Pankaj Sharma
2.	Web Technology: A Developer's Perspective (Paperback) by N.P. Gopalan

Reference Books	
1.	Web Technology Theory and Practice By Akshi Kumar
2.	Introduction to Web Interaction Design With HTML and CSS By Michael Macaulay

Useful Links	
1.	https://www.atri.edu.in/images/pdf/departments/Web%20Technologies%20Lab%20Manual.pdf
2.	https://www.lendi.org/CSE/labmanuals/Web%20Technologies.pdf

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Program: B. Tech. in Computer Science & Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
III	**CS3F001**	**Community Engagement Project**	0	0	4	2

Prerequisites for the course	
1.	Understand the visiting companies shall be relevant and suitable ones to the specialization and academic requirements.
2	Learning Important tool for attracting investment and technology.

Prior Reading Material/useful links	
1.	https://www.quora.com/What-are-some-good-places-for-an-industrial-trip-of-CSE-students
2.	https://www.suas.ac.in/industrial-visit-for-computer-science-and-information-technology-5th-sem-students/

Course Outcomes:

Sr. No	Course Outcome number	CO statement
1	CO1	Students will able to functional opportunity in different sectors.
2	CO2	Students will able to helps to combine theoretical knowledge with industrial knowledge.
3	CO3	Students will able to provide an insight into the real working environment,
4	CO4	Students will able to get to learn a lot of things that will help in their development and also for the future.
5	CO5	Students will able to acquire and apply fundamental principles of science and engineering.

Community engagement is based on the democratic idea that everyone who is affected by an issue that impacts their community should have a say in the decision making around it. It, moreover, holds the promise that public participation can influence decisions that affect the provision of services, future visions and sustainability of our communities.

At its most basic level, community engagement is striving towards improving the quality of life in a given community through both political and non-political means. Community engagement is also commonly referred to as civic engagement or public participation, and all terms reference a desire to make a positive difference in one's community that manifests itself through action, combining knowledge, skills, values and motivation.

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Although voting often comes to mind when people discuss the topic of community engagement, it's important to know that it's much more than that. In fact, voting could be considered the bare minimum of community engagement. While voting is certainly a civic duty, or responsibility, engaging in one's community should never stop there.

Examples of Community Engagement

Among the many different forms of community engagement, here are just a few:

- Volunteering at a local food bank, animal shelter or homeless shelter
- Maintaining a community garden
- Participating in a local blood drive
- Helping set up a local farmers market
- Advocating for others
- Creating support groups in your community
- Attending and participating in town hall and city council meetings
- Attending and participating in school board meetings
- Voting in local elections

Models for Community Engagement

Along with the many forms of community engagement, there are also many different models for it. These models can include relationship-building projects that help bring a community together and form stronger bonds; community education projects that provide instructional services or educate the public on certain issues; direct service projects that provide something to a group in need; and more. These varying forms of community engagement happen in communities all across the globe. While not every community has the same level of engagement, the same percentage of the population that's passionate about it, or the same forms of community engagement, it shouldn't be a surprise to learn that the ones with higher levels of engagement fare better.

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Sr. No.	Course Code	Course Name	Subject In charge Name
1	CS3T001	Computer Networks	Prof. Anuja Ghasad
2	CS3T002	Operating Systems	Prof. Swati Raut
3	CS3T003	Computer Architecture & Organization	Prof. Kiran Bode
4	CS3M001	Numerical Technique & Discrete Mathematics	Prof. Ashwini Ramteke
5	CS3O001	Open Elective -I	Prof. Chetana Khandale
6	CS3H001	Entrepreneurship Development	Prof. Siddharth Ghosh
7	CS3V001	UHV-II	Prof. Mittal Patne
8	CS3L005	Object Techniques & UML Lab	Ms. Gayatri Gawande
9	CS3L006	Python Programming Lab	Ms. Rashmi Pathan
10	CS3L007	Web Technology Lab	Ms. Jagruti Thakre
11	CS3F001	Community Engagement Project	

Prof. Anuja Ghasad,
Secretary BOS, CSE,
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